

Euler's Totient Function

Euler's totient function, $\phi(n)$, gives the number of positive integers less than or equal to n that are relatively prime to n .

Examples:

$$\phi(7) = 6$$

$$\phi(10) = 4$$

$$\phi(12) = 4$$

$$\phi(16) = 8$$

$$\phi(29) = 28$$

Special cases:

- When n is a prime number, p , then $\phi(p) = p - 1$.
- When $n = p \times q$ with p and q both primes, we can compute $\phi(n)$ by subtracting the number of integers that are NOT relatively prime to n from n . In other words, subtract all p multiples of q , subtract all q multiples of p ... and then add one back because pq was taken away twice:
$$\begin{aligned}\phi(n) &= n - p - q + 1 \\ &= pq - p - q + 1 \\ &= (p-1)(q-1)\end{aligned}$$

Examples:

$$\begin{aligned}\phi(10) &= 10 - 5 - 2 + 1 \\ &= (5)(2) - 5 - 2 + 1 \\ &= (5-1)(2-1) \\ &= 4\end{aligned}$$

How many integers are relatively prime to 2537, given that $2537 = 43 \times 59$?

Fermat's Little Theorem

Fermat's little theorem states that a^{p-1} is congruent to 1 modulus p , where p is a prime and a is not a multiple of p :

$$a^{p-1} \equiv 1 \pmod{p}$$

We can use this theorem to determine an inverse to a modulo p (but only when p is prime!).

$$a^{p-1} \equiv 1 \pmod{p}$$

$$(a^{p-2})a \equiv 1 \pmod{p}$$

The inverse to a modulo p is a^{p-2} .

Example: Find an inverse to 6 modulo 17.

17 is a prime so we can use Fermat's little theorem: $(6^{17-2})6 \equiv 1 \pmod{17}$.

So we need to compute $6^{17-2} = 6^{15} \pmod{17}$ using fast modular exponentiation:

$$6^2 \pmod{17} = 2$$

$$6^4 \pmod{17} = 4$$

$$6^8 \pmod{17} = 16$$

$$6^{8+4+2+1=15} \pmod{17} = (16)(4)(2)(6) \pmod{17} = 768 \pmod{17} = 3 \pmod{17}$$

So 3 is an inverse of 6 modulo 17 because $(3)6 \equiv 1 \pmod{17}$

Note: The Extended Euclidean Algorithm also gives 3 as an inverse:

	Quotient	Remainder	s	t
$17 \bmod 6 = 5$	2	$(17)-2(6)=5$	1	-2
$6 \bmod 5 = 1$	1	$(6)-1(5)=1$	-1	$1-1(-2)=3$
$5 \bmod 1 = 0$				

We have that $\text{GCD}(17,6)=1$, and $-1(17)+3(6)=1$

So 3 is an inverse of 6 modulo 17 because $(3)6 \equiv 1 \pmod{17}$

Euler's Theorem

Euler's Theorem states that for an integer a relatively prime to n , $a^{\phi(n)}$ is congruent to 1 modulo n :

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

When $n = p \times q$, with p and q both primes:

$$a^{(p-1)(q-1)} \equiv 1 \pmod{n}$$

Note that Euler's theorem applies to any integer n (prime or composite). When we make n a prime (let $n=p$ where p is prime) we have:

$$a^{\phi(p)} \equiv 1 \pmod{p}$$

$$a^{p-1} \equiv 1 \pmod{p}$$

Which is Fermat's Little Theorem.

Euler's Theorem and RSA

Recall the RSA cryptosystem where $n = p \times q$, with p and q both primes and we compute the inverse of $e \pmod{(p-1)(q-1)}$ to get the decryption exponent d such that $ed \equiv 1 \pmod{(p-1)(q-1)}$.

Then: $ed = 1 + r((p-1)(q-1))$, for some integer r .

The encrypted message m^e is decrypted by computing $(m^e)^d$:

$$\begin{aligned} (m^e)^d &= m^{ed} \pmod{n} \\ &= m^{1+r(p-1)(q-1)} \pmod{n} \\ &= m^1 m^{r(p-1)(q-1)} \pmod{n} \\ &= m^1 m^{r\phi(n)} \pmod{n} \\ &= m^1 \left(m^{\phi(n)} \right)^r \pmod{n} \\ &= m^1 (1)^r \pmod{n} \\ &= m \pmod{n} \end{aligned}$$

Example: Find an inverse to 71 modulo 481, where $(13)(37)=481$.

481 is composite (a product of two primes) so we cannot use Fermat's little theorem.

Using Euler's theorem:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

$$71^{\phi(481)} \equiv 1 \pmod{481}$$

$$71^{(13-1)(37-1)} \equiv 1 \pmod{481}$$

$$71^{(12)(36)} \equiv 1 \pmod{481}$$

$$71^{432} \equiv 1 \pmod{481}$$

$$71^{431} 71 \equiv 1 \pmod{481}$$

Using fast modular exponentiation we find that $71^{431} \pmod{481} = 271$.

So 271 is an inverse of 71 modulo 481 because $(271)71 \equiv 1 \pmod{481}$

Alternatively we could use the force (of the Extended Euclidean Algorithm).

	Quotient	Remainder	s	t
$481 \bmod 71 = 55$	6	55	1	-6
$71 \bmod 55 = 16$	1	16	-1	$1 - 1(-6) = 7$
$55 \bmod 16 = 7$	3	7	$1 - 3(-1) = 4$	$-6 - 3(7) = -27$
$16 \bmod 7 = 2$	2	2	$-1 - 2(4) = -9$	$7 - 2(-27) = 61$
$7 \bmod 2 = 1$	3	1	$4 - 3(-9) = 31$	$-27 - 3(61) = -210$
$2 \bmod 1 = 0$				

We have that $\text{GCD}(481, 71) = 1$ and $31(481) - 210(71) = 1$.

-210 is congruent to 271 modulo 481, because $-210 + 481 = 271$.

So 271 is an inverse of 71 modulo 481.

Exercises:

1. Determine the number of integers that are relatively prime to:
 - a) 14
 - b) 713
2. What are the values of n for which $\phi(n) = 2$?
3. How many possible private keys, $\{a, b\}$, are there for the affine cipher:
 - a) $f(M) = (aM + b) \bmod 10$
 - b) $f(M) = (aM + b) \bmod 26$
 - c) $f(M) = (aM + b) \bmod 23$
4. Use Fermat's little theorem to find an inverse to:
 - a) 17 modulo 37
 - b) 17 modulo 1069
5. Find an inverse to 17 modulo 713.

Answers:

1.

- a) 6
- b) 660

2. 3, 4, 6

3.

- a) 40
- b) 312
- c) 506

4.

- a) 24
- b) 566

5. The answer to life the universe and everything (42).